

Retrieval-based forecast adaptation

Executive summary of results

Adaptation pipeline and notations

For query lookback X and target $T \in \mathbb{R}^H$, the frozen backbone F produces the vanilla forecast $V = F(X)$. Retrieval returns K historical pairs (X_k, Y_k) with distance weights w_k ; their backbone forecasts are $N_k = F(X_k)$, their weighted future is $\bar{Y} = \sum_{k=1}^K w_k Y_k$, and the retrieved pairs also produce the covariate-conditioned forecast $C = F(X; \{(X_k, Y_k)\}_{k=1}^K)$. The adaptor maps $(V, C, Y_{1:K}, N_{1:K}, \bar{Y})$ to the final forecast $\hat{T}$. For a candidate adaptation A , its realized advantage over vanilla is

$$\Delta(A) = \frac{1}{H} \sum_{h=1}^H \left[(T_h - V_h)^2 - (T_h - A_h)^2 \right].$$

We propose various *baselines* corresponding to different formulas fitted on $C = F(X; \{(X_k, Y_k)\}_{k=1}^K)$ to predict T , as well as *gates* which take binary decisions to choose between V and a given non-trainable adaptor $A \in \{C, \bar{Y}\}$.

Practical details

Component	Protocol
Datasets	Electricity, Traffic, Solar, and <code>exchange_rate</code> .
Settings	$(L, H) \in \{(168, 24), (336, 48), (504, 168)\}$: 12 dataset-setting configurations.
Backbone	Frozen <code>chronos2</code> .
Splits	T_0 datastore (30%); T_1 training (40%); T_2 validation (20%), T_3 eval (20%).

For a given configuration c and method m , we consider the mean relative improvement in nMSE over vanilla v :

$$I_{m,c} = 100 \frac{\text{nMSE}_{m,c} - \text{nMSE}_{m,v}}{\text{nMSE}_{m,v}} \%.$$

Tables report the mean of $I_{m,c}$ over the 12 held-out T_3 configurations.

Selected baseline and gate candidates

Code name	Forecast formula	Space	K	T3 gain
<code>full_ridge_shared</code>	$V_h + \beta_V V_h + \beta_C C_h + \sum_k (\beta_k^Y Y_{k,h} + \beta_k^N N_{k,h})$	instance	3	+2.733%
<code>y_ridge_shared</code>	$V_h + \beta_V V_h + \sum_k \beta_k^Y Y_{k,h}$	instance	3	+2.293%
<code>cov_y_ridge_shared</code>	$V_h + \beta_V V_h + \beta_C C_h + \sum_k \beta_k^Y Y_{k,h}$	instance	3	+2.546%
<code>residual_ridge_shared</code>	$V_h + \beta_V V_h + \sum_k (\beta_k^Y Y_{k,h} + \beta_k^N N_{k,h})$	instance	3	+2.539%
<code>cov_avgy_ridge_shared</code>	$V_h + \beta_V V_h + \beta_C C_h + \beta_{\bar{Y}} \bar{Y}_h$	raw	3	+2.306%
<code>cov_ridge_shared</code>	$V_h + \beta_V V_h + \beta_C C_h$	raw	3	+2.142%
<code>catboost_avgy_regressor_shared</code>	$G(\bar{Y}; g_\theta(z))_h$	instance	3	+0.019%
<code>catboost_cov_regressor_shared</code>	$G(C; g_\theta(z))_h$	instance	1	+0.015%
<code>bayes_cov_shared</code>	$G(C; b(C))_h$	instance	1	-0.003%

All six ridge candidates improve the mean held-out nMSE, whereas the three selected gates are effectively tied with vanilla. The strongest candidate at its screened configuration is `full_ridge_shared` at +2.733%.

Formulation ablations

Each row holds the variable design and retrieval pipeline fixed. The Δ -ridge and convex code names replace `_ridge_shared` by `_delta_ridge_shared` and `_convex_shared`, respectively.

Shared-ridge control (code name)	Ridge	Δ -ridge	Convex
<code>full_ridge_shared</code>	2.733%	1.515%	0.635%
<code>y_ridge_shared</code>	2.293%	1.347%	1.209%
<code>cov_y_ridge_shared</code>	2.546%	1.468%	1.115%
<code>residual_ridge_shared</code>	2.539%	1.339%	0.330%
<code>cov_avgy_ridge_shared</code>	2.306%	1.183%	1.063%
<code>cov_ridge_shared</code>	2.142%	0.652%	0.347%

Ordinary shared ridge is best for every matched design.

Effect of the neighbour count K

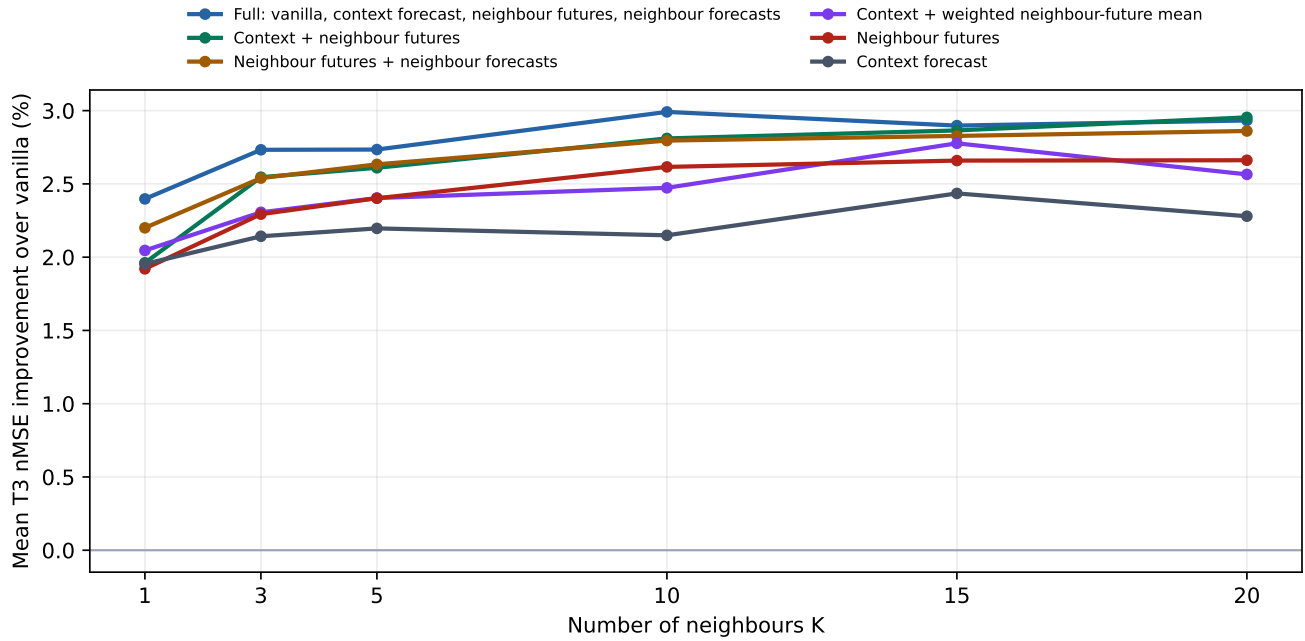


Figure 1: Mean held-out T3 nMSE improvement over 12 configurations. Only the six selected ridge baselines are plotted.

Candidate (code name)	Family	Space	Best K	T3 gain
<code>full_ridge_shared</code>	baseline	instance	10	+2.991%
<code>y_ridge_shared</code>	baseline	instance	20	+2.661%
<code>cov_y_ridge_shared</code>	baseline	instance	20	+2.954%
<code>residual_ridge_shared</code>	baseline	instance	20	+2.861%
<code>cov_avgy_ridge_shared</code>	baseline	raw	15	+2.777%
<code>cov_ridge_shared</code>	baseline	raw	15	+2.435%
<code>catboost_avgy_regressor_shared</code>	gate	instance	3	+0.019%
<code>catboost_cov_regressor_shared</code>	gate	instance	1	+0.015%
<code>bayes_cov_shared</code>	gate	instance	1	-0.003%

The ridge candidates peak between $K = 10$ and $K = 20$; the gates peak at their screen values and remain negligible. The unfinished $K = 100$ extension is not included.

TS-IFA results

Code name	Space / steps	T2 gain	T3 gain	Checkpoint / held-out behaviour
joint_ridge_horizon_unconstrained_full	raw / 10K	+1.84%	+0.04%	Window gains +3.26, +0.58, −3.72%; all three select 10K.
joint_ridge_horizon_unconstrained_full	raw / 30K	+3.03%	−6.71%	Selected at 29/20/20K for (168/24, 336/48, 504/168).
joint_ridge_horizon_unconstrained_full	instance / 30K	+3.25%	−17.97%	Selected at 30/26/20K for the same windows.
joint_neural_horizon_unconstrained_full	raw / instance, 30K	−	−4.60/ −2.14%	Zero T3 wins.

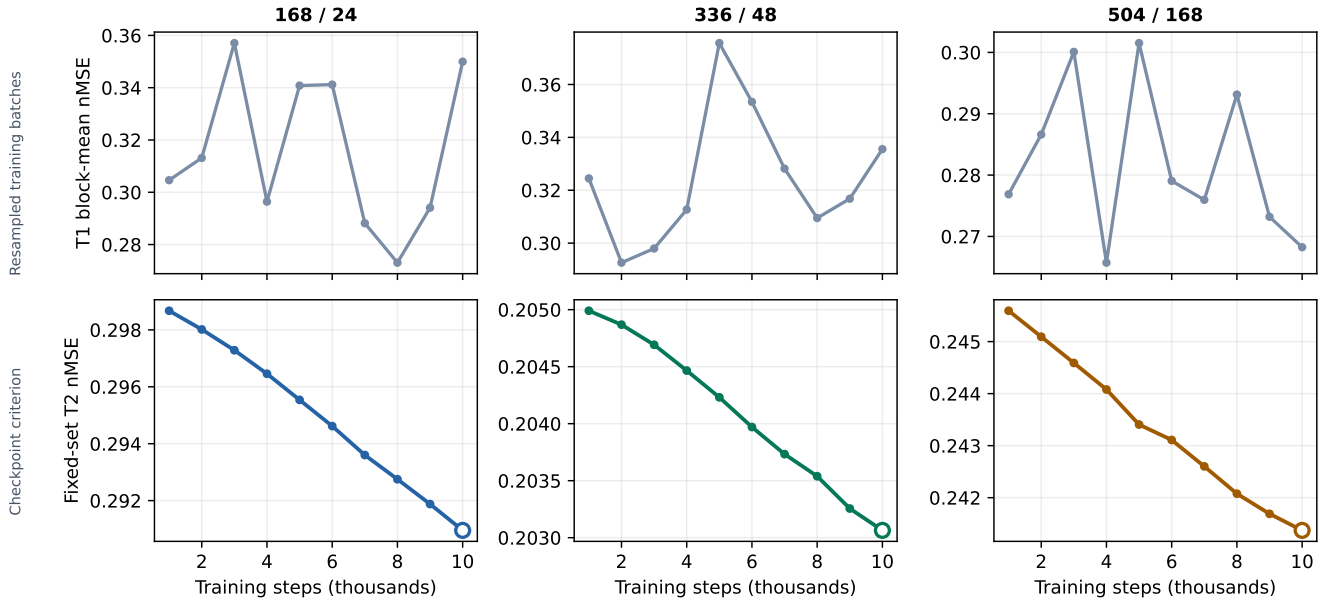


Figure 2: The 10K joint-ridge criterion curves. The gray T1 trace is a 1,000-step block mean over freshly resampled training batches, not a repeated evaluation on one fixed set, so it need not be monotone. The colored T2 trace uses a fixed set and selects 10K for all three windows.

TS-IFA is not reliable on held-out T3. Longer runs improve T2 selection but worsen T3, with the largest failures at $H = 168$; the raw 10K ridge pilot remains the least-bad aggregate result at +0.04%.